

23CSE472 – AI & Robotics Lab

3D Rotation Representation

1. Transformation

`PLOT_POINT(P, OPTIONS)` adds point markers and optional annotation text to the current plot, where P ($2 \times N$) and each column is a point coordinate.
`H = PLOT_POINT(...)` as above but return handles to the points.

$H = E2H(E)$ is the homogeneous version ($K + 1 \times N$) of the Euclidean points E ($K \times N$) where each column represents one point in R^K .

$E = H2E(H)$ is the Euclidean version ($K - 1 \times N$) of the homogeneous points H ($K \times N$) where each column represents one point in P^K .

```
>> B = transl2(1,2)
```

```
B =
```

```
1    0    1
0    1    2
0    0    1
```

```
>> AP = [2;3]
```

```
AP =
```

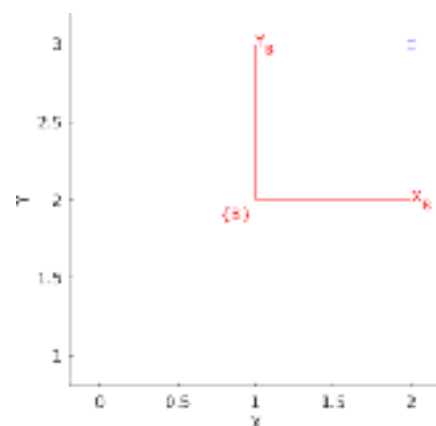
```
2
```

```
3
```

```
>> trplot2(B, 'frame','B', 'color','r')
```

```
>> hold on
```

```
>> plot_point(AP)
```



```

>> PB = inv(B) * [AP;1]

PB =

     1
     1
     1

>> h2e(PB)

ans =

     1
     1

>> PB = h2e(inv(B) * e2h(AP))

PB =

     1
     1

```

2. Symbolic mathematics also possible. (Symbolic Math Toolbox is required)

```

>> syms theta

>> R = rot2(theta)

R =

 [ cos(theta), -sin(theta)]
 [ sin(theta),  cos(theta)]

>> simplify (R*R)

ans =

 [ cos(2*theta), -sin(2*theta)]
 [ sin(2*theta),  cos(2*theta)]

```

Note: Try subs and eval

3. Matrix Exponential

$V = \text{VEX}(S)$ is the vector which has the corresponding skew-symmetric matrix S .

In the case that $S (2 \times 2) =$

$$\begin{vmatrix} 0 & -v \\ v & 0 \end{vmatrix}$$

$$\begin{vmatrix} v & 0 \\ 0 & 0 \end{vmatrix}$$

then $V = [v]$. In the case that $S (3 \times 3) =$

$$\begin{vmatrix} 0 & -yz & vx \\ yz & 0 & vy \\ -vy & vx & 0 \end{vmatrix}$$

$$\begin{vmatrix} vx & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$\begin{vmatrix} -vy & vx & 0 \\ vx & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$S = \text{SKEW}(V)$ is a skew-symmetric matrix formed from V .

```
>> R = rot2(45, 'deg')  
  
R =  
  
    0.7071    -0.7071  
    0.7071     0.7071  
  
>> S = logm(R)  
  
S =  
  
     0    -0.7854  
    0.7854     0  
  
>> vex(S)  
  
ans =  
  
    0.7854  
  
>> skew(ans)  
  
ans =  
  
     0    -0.7854  
    0.7854     0  
  
>> expm(S)  
  
ans =  
  
    0.7071    -0.7071  
    0.7071     0.7071
```

4. Alternate Representations

$R = \text{EUL2R}(\text{PHI}, \text{THETA}, \text{PSI}, \text{OPTIONS})$ is an $\text{SO}(3)$ orthonormal rotation matrix (3×3) equivalent to the specified Euler angles. These correspond to rotations about the Z, Y, Z axes respectively.

$T = \text{EUL2TR}(\text{PHI}, \text{THETA}, \text{PSI}, \text{OPTIONS})$ is an $\text{SE}(3)$ homogeneous transformation matrix (4×4) with zero translation and rotation equivalent to the specified Euler angles. These correspond to rotations about the Z, Y, Z axes respectively.

$\text{EUL} = \text{TR2EUL}(T, \text{OPTIONS})$ are the ZYZ Euler angles (1×3) corresponding to the rotational part of a homogeneous transform T (4×4). The 3 angles

$\text{EUL} = [\text{PHI}, \text{THETA}, \text{PSI}]$

correspond to sequential rotations about the Z, Y and Z axes respectively.

$\text{EUL} = \text{TR2EUL}(R, \text{OPTIONS})$ as above but the input is an orthonormal rotation matrix R (3×3).

Check out: TR2RPY , RPY2TR , OA2R , OA2TR , TR2ANGVEC , ANGVEC2R , ANGVEC2TR (refer all these functions in toolbox manual)

```
>> R = eul2r(0.1,0.2,0.3)

R =

    0.9021    -0.3836     0.1977
    0.3875     0.9216     0.0198
   -0.1898     0.0587     0.9801

>> EU= tr2eul(R)

EU =

    0.1000     0.2000     0.3000
```

Mapping from rotation matrix to Euler angle is not unique

```
>> R = eul2r(0.1,-0.2,0.3)

R =

    0.9021    -0.3836    -0.1977
    0.3875     0.9216    -0.0198
    0.1898    -0.0587     0.9801
```

```

>> EU = tr2eul(R)

EU =
    -3.0416    0.2000   -2.8416

>> R1 = eul2r(EU)

R1 =
    0.9021   -0.3836   -0.1977
    0.3875    0.9216   -0.0198
    0.1898   -0.0587    0.9801

```

Try Out

1. Create a Homogeneous transformation matrix H for a frame with translation (1,4,8), rotation about $x - 45\text{deg}$, rotation about $y-70\text{deg}$ and rotation about $z-30\text{deg}$.
 - a. Convert H into RPY angles
 - b. Convert H into angle vector representation
 - c. Convert H into euler angles